Curriculum Learning for Time-Series Users

Problem formulation and proposed treatment

Thesis project

Standalone method note

Abstract

This note isolates the scientific formulation from implementation and cluster operations. It is intended to be readable on its own and reusable when drafting a paper introduction or method section.

1 Forecasting task

Let $x_u(1{:}T)$ be the series of user $u \in \mathcal{U}$. At query time t, the model receives $X_{u,t} = x_u(t - L + 1{:}t) \in \mathbb{R}^L$ and predicts $Y_{u,t} = x_u(t + 1{:}t + H) \in \mathbb{R}^H$. Training and evaluation splits are chronological and the complete target horizon belongs to its target split.

2 Causal user difficulty

Difficulty is computed from accessible training windows only. Persistence uses $\widehat{Y}_{u,t,h} = x_u(t)$ and gives the normalized window error

$$d_{u,t} = \frac{\mathrm{MSE}(Y_{u,t}, \widehat{Y}_{u,t})}{(\mathrm{std}(X_{u,t}) + \varepsilon)^2}.$$

The default user score is $d_u = \mathrm{median}_t\, d_{u,t}$. Users are sorted by (d_u, u), using the stable source identity to resolve ties.

3 Proposed curriculum

Training has K phases and a fixed total of S optimizer steps. Phase k activates a cumulative prefix $A_k \subseteq \mathcal{U}$ of the ordered users. At step s in phase k, the sampler draws

$$u_s \sim \mathrm{Uniform}(A_k), \qquad (X_{u_s,t_s}, Y_{u_s,t_s}) \sim \mathcal{W}^{\mathrm{train}}_{u_s}.$$

The model, optimizer, scheduler, and global step persist across phase boundaries. Only the user sampling distribution changes.

4 Controls and interpretation

Uniform sampling tests the value of any curriculum. Hard-to-easy and a seeded random order test whether the chosen order matters. The exposure-matched control removes temporal order while retaining each user's expected number of updates:

$$E_u = \sum_{k=1}^{K} S_k \frac{\mathbf{1}[u \in A_k]}{|A_k|}, \qquad p_u = \frac{E_u}{\sum_v E_v}.$$

A gain over uniform but not over exposure matching indicates reweighting; a gain over both supports a temporal-order effect.

5 Reported evidence

All methods share backbone, initialization distribution, optimizer budget, batch size, loss, and evaluation rows. Reports include global error, equal-user error, worst-10-percent-user error, convergence by optimizer step, and error by difficulty quantile.